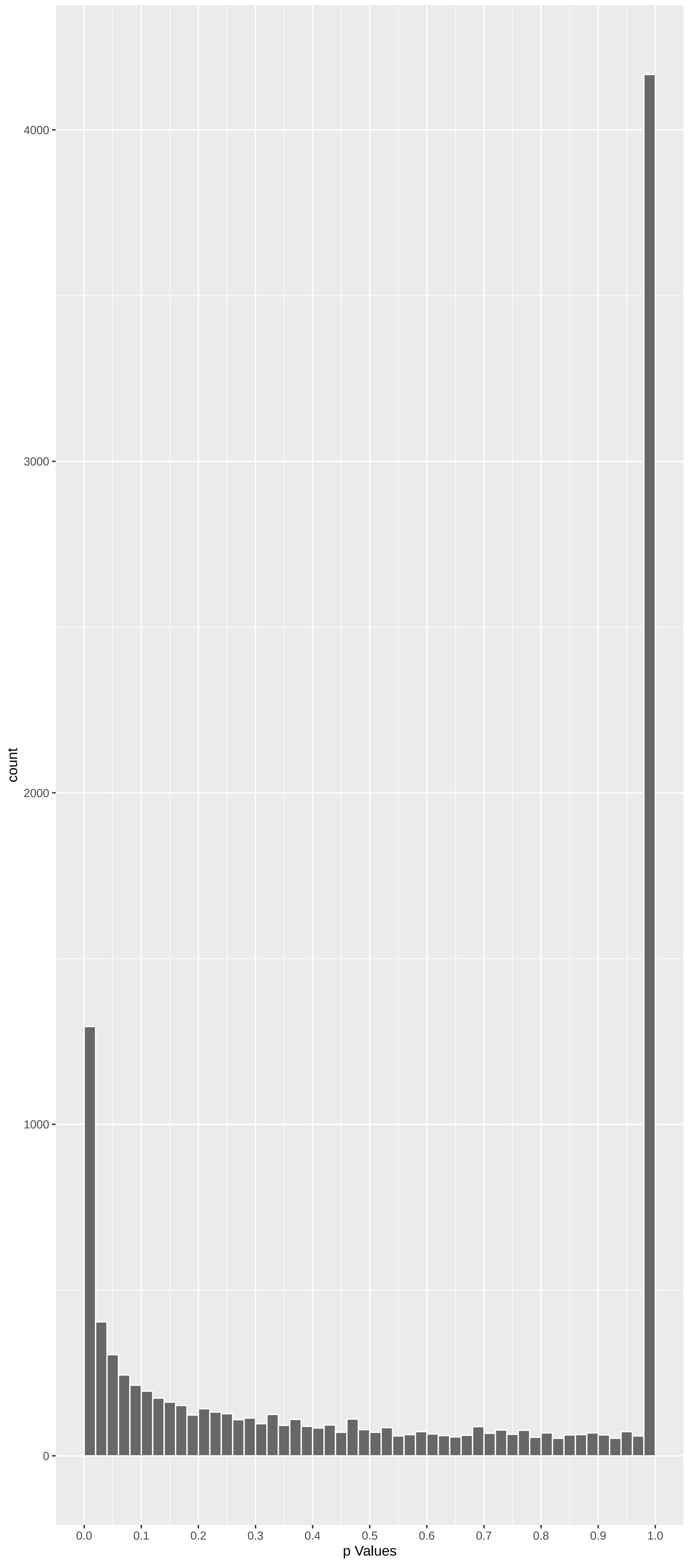
Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
TRIM15	−7.063450	4.872546e−12	7.859e−08	1.268e−03
SPEGNB	6.627997	1.020819e−10	8.233e−07	6.640e−03
DNAH14	−6.322894	7.701305e−10	4.141e−06	1.207e−02
FGD5	6.225750	1.437769e−09	5.798e−06	1.207e−02
BIRC3	−6.180452	1.917552e−09	6.186e−06	1.207e−02
LDLRAD4	6.142569	2.435921e−09	6.549e−06	1.207e−02
CNTF	6.068090	3.883205e−09	6.960e−06	1.207e−02
DCAF12L1	6.072901	3.768600e−09	6.960e−06	1.207e−02
FAR2	6.101110	3.160041e−09	6.960e−06	1.207e−02
BTBD18	6.018804	5.271316e−09	8.503e−06	1.207e−02
TTC9C	5.980391	6.678086e−09	8.976e−06	1.207e−02
FBXO42	5.983438	6.554292e−09	8.976e−06	1.207e−02
HIRA	5.831901	1.643981e−08	2.040e−05	2.531e−02
CTNND1	5.803606	1.947107e−08	2.243e−05	2.585e−02
NUP160	5.773949	2.323045e−08	2.498e−05	2.641e−02
GPR179	−5.755023	2.598903e−08	2.620e−05	2.641e−02
ZFAND4	5.701441	3.563966e−08	3.382e−05	3.082e−02
ATP6V0A2	5.688775	3.838628e−08	3.440e−05	3.082e−02
GYS2	5.663361	4.453085e−08	3.780e−05	3.209e−02
DYNC1LI1	5.548748	8.631669e−08	6.784e−05	5.013e−02
ZNF311	−5.544739	8.831764e−08	6.784e−05	5.013e−02
RBSN	5.524685	9.902293e−08	7.148e−05	5.013e−02
ATP10A	5.519605	1.019290e−07	7.148e−05	5.013e−02
RFX8	5.493083	1.184928e−07	7.964e−05	5.352e−02
SLC13A5	−5.456626	1.455798e−07	9.393e−05	6.060e−02

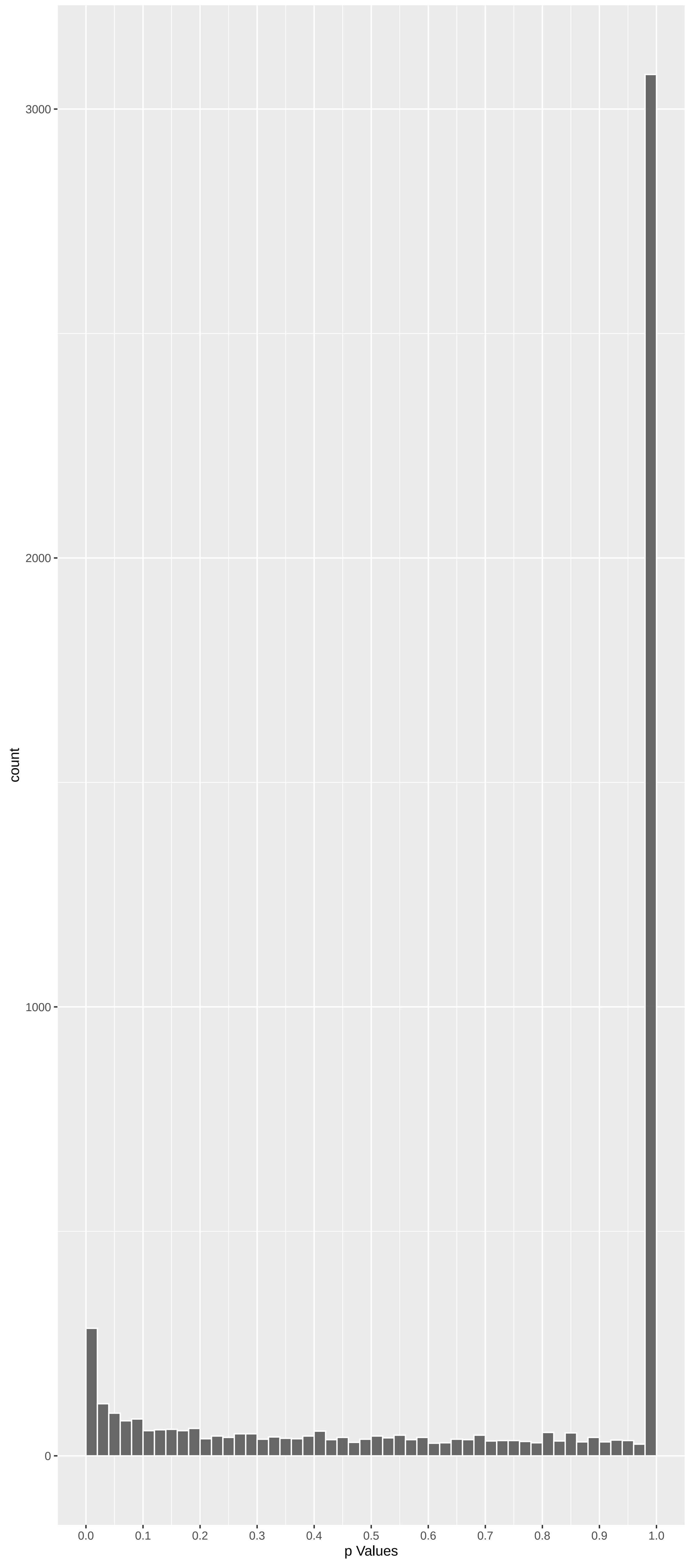
Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
TRIM15	−7.063450	4.872546e−12	7.859e−08	1.268e−03
SPEGNB	6.627997	1.020819e−10	8.233e−07	6.640e−03
TTC9C	5.980391	6.678086e−09	8.976e−06	1.207e−02
LDLRAD4	6.142569	2.435921e−09	6.549e−06	1.207e−02
CNTF	6.068090	3.883205e−09	6.960e−06	1.207e−02
DCAF12L1	6.072901	3.768600e−09	6.960e−06	1.207e−02
FBXO42	5.983438	6.554292e−09	8.976e−06	1.207e−02
BTBD18	6.018804	5.271316e−09	8.503e−06	1.207e−02
DNAH14	−6.322894	7.701305e−10	4.141e−06	1.207e−02
FAR2	6.101110	3.160041e−09	6.960e−06	1.207e−02
FGD5	6.225750	1.437769e−09	5.798e−06	1.207e−02
BIRC3	−6.180452	1.917552e−09	6.186e−06	1.207e−02
HIRA	5.831901	1.643981e−08	2.040e−05	2.531e−02
CTNND1	5.803606	1.947107e−08	2.243e−05	2.585e−02
NUP160	5.773949	2.323045e−08	2.498e−05	2.641e−02
GPR179	−5.755023	2.598903e−08	2.620e−05	2.641e−02
ATP6V0A2	5.688775	3.838628e−08	3.440e−05	3.082e−02
ZFAND4	5.701441	3.563966e−08	3.382e−05	3.082e−02
GYS2	5.663361	4.453085e−08	3.780e−05	3.209e−02
RBSN	5.524685	9.902293e−08	7.148e−05	5.013e−02
DYNC1LI1	5.548748	8.631669e−08	6.784e−05	5.013e−02
ATP10A	5.519605	1.019290e−07	7.148e−05	5.013e−02
ZNF311	−5.544739	8.831764e−08	6.784e−05	5.013e−02
RFX8	5.493083	1.184928e−07	7.964e−05	5.352e−02
SLC13A5	−5.456626	1.455798e−07	9.393e−05	6.060e−02

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
SPEGNB	6.627997	1.020819e−10	8.233e−07	6.640e−03
FGD5	6.225750	1.437769e−09	5.798e−06	1.207e−02
LDLRAD4	6.142569	2.435921e−09	6.549e−06	1.207e−02
CNTF	6.068090	3.883205e−09	6.960e−06	1.207e−02
DCAF12L1	6.072901	3.768600e−09	6.960e−06	1.207e−02
FAR2	6.101110	3.160041e−09	6.960e−06	1.207e−02
BTBD18	6.018804	5.271316e−09	8.503e−06	1.207e−02
TTC9C	5.980391	6.678086e−09	8.976e−06	1.207e−02
FBXO42	5.983438	6.554292e−09	8.976e−06	1.207e−02
HIRA	5.831901	1.643981e−08	2.040e−05	2.531e−02

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
TRIM15	−7.063450	4.872546e−12	7.859e−08	1.268e−03
DNAH14	−6.322894	7.701305e−10	4.141e−06	1.207e−02
BIRC3	−6.180452	1.917552e−09	6.186e−06	1.207e−02
GPR179	−5.755023	2.598903e−08	2.620e−05	2.641e−02
ZNF311	−5.544739	8.831764e−08	6.784e−05	5.013e−02
SLC13A5	−5.456626	1.455798e−07	9.393e−05	6.060e−02
PIP	−5.313418	3.227635e−07	1.735e−04	9.331e−02
PLPP1	−5.115388	9.392892e−07	3.632e−04	1.395e−01
N4BP2L2	−5.118147	9.256550e−07	3.632e−04	1.395e−01
SCNN1D	−5.031394	1.460780e−06	4.909e−04	1.650e−01

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO:	−0.22510823	48	6.965e−08	1.251e−04	NDUFA11:162 NDUFB10:273 TMEM126B:418 NDUFA8:467 NDUFB8:482 OXA1L:507
Inorganic Cation Transmembrane Transport	0.09543290	277	5.296e−08	1.251e−04	ATP6V0A2:14 ATP2A1:29 KCNK18:32 KCNA4:111 SLC6A6:123 TRPM7:128
Mitochondrial Respiratory Chain Complex	−0.22510823	48	6.965e−08	1.251e−04	NDUFA11:162 NDUFB10:273 TMEM126B:418 NDUFA8:467 NDUFB8:482 OXA1L:507
Monoatomic Cation Transmembrane Transpor	0.08795072	273	6.393e−07	8.612e−04	ATP6V0A2:14 ATP2A1:29 KCNK18:32 KCNA4:111 SLC6A6:123 TRPM7:128
Oxidative Phosphorylation (GO:0006119)	−0.18781018	55	1.478e−06	1.403e−03	ATP5PD:95 NDUFA11:162 NDUFB10:273 NDUFV3:343 ATP5MG:452 NDUFA8:467
Proton Motive Force−Driven Mitochondrial	−0.20263110	47	1.563e−06	1.403e−03	ATP5PD:95 NDUFA11:162 NDUFB10:273 NDUFV3:343 ATP5MG:452 NDUFA8:467
Monocarboxylic Acid Transport (GO:001571	0.17315845	63	2.045e−06	1.574e−03	SLC27A4:52 TSPO2:116 ATP8B1:130 SLC10A2:245 MFSD2A:342 SLC5A8:368
Peptide Biosynthetic Process (GO:0043043	−0.11756800	134	2.744e−06	1.848e−03	MRPL24:144 GGT6:259 MRPL37:313 CARNS1:329 RPS8:498 PABPC4:499
Mitochondrial Respiratory Chain Complex	−0.15267363	77	3.719e−06	2.227e−03	NDUFA11:162 NDUFB10:273 LYRM7:352 TMEM126B:418 NDUFA8:467 NDUFB8:482
Defense Response To Bacterium (GO:004274	−0.12360656	115	4.863e−06	2.620e−03	NLRP1:41 DEFB132:147 TLR6:316 IL18:356 GSDMC:533 AZU1:572
Proton Motive Force−Driven ATP Synthesis	−0.18179446	51	7.188e−06	3.521e−03	ATP5PD:95 NDUFA11:162 NDUFB10:273 NDUFV3:343 ATP5MG:452 NDUFA8:467
Carboxylic Acid Transport (GO:0046942)	0.18057216	50	1.014e−05	4.553e−03	SLC43A1:48 SLC6A6:123 SLC6A15:190 SLC36A2:369 SLC6A19:389 SLC6A14:604
Cilium−Dependent Cell Motility (GO:00602	−0.30659579	17	1.209e−05	5.011e−03	DNAH2:29 TEKT1:449 DNAH3:458 CFAP54:564 DNAH8:726 DNAH1:734
Potassium Ion Transmembrane Transport (G	0.10844812	132	1.760e−05	6.774e−03	KCNK18:32 KCNJ16:53 KCNA4:111 KCNQ5:131 KCNJ1:191 LRRC55:204
Nitrogen Compound Transport (GO:0071705)	0.10023472	151	2.226e−05	7.995e−03	SLC43A1:48 SLC6A6:123 SLC22A4:152 RD3:183 SLC6A15:190 SLC22A1:300
Axoneme Assembly (GO:0035082)	−0.22230355	30	2.523e−05	8.497e−03	CCDC40:36 TTC12:298 RP1:513 RSPH6A:649 SPAG1:795 CFAP74:880
Aerobic Respiration (GO:0009060)	−0.16225588	54	3.767e−05	1.194e−02	NDUFA11:162 NDUFB10:273 NDUFV3:343 NDUFA8:467 NDUFB8:482 OXA1L:507
Mitochondrial Electron Transport, NADH T	−0.21157163	31	4.592e−05	1.375e−02	NDUFB10:273 NDUFV3:343 NDUFA8:467 NDUFB8:482 NDUFS5:745 NDUFS7:1569
Fatty Acid Transport (GO:0015908)	0.18985400	37	6.485e−05	1.839e−02	SLC27A4:52 CD36:159 SLC22A1:300 FABP9:334 MFSD2A:342 FABP1:438
Cellular Respiration (GO:0045333)	−0.13453245	71	9.017e−05	2.429e−02	NDUFA11:162 NDUFB10:273 NDUFV3:343 LYRM7:352 NDUFA8:467 NDUFB8:482
Mitochondrial Gene Expression (GO:014005	−0.10967197	100	1.544e−04	3.961e−02	POLRMT:42 DAP3:105 MRPL24:144 MRPL37:313 FASTKD5:469 HARS1:767
Mitochondrial ATP Synthesis Coupled Elec	−0.14641987	55	1.746e−04	4.275e−02	NDUFB10:273 NDUFV3:343 NDUFA8:467 NDUFB8:482 NDUFA12:670 NDUFS5:745
Translation (GO:0006412)	−0.07551397	206	1.965e−04	4.602e−02	DAP3:105 MRPL24:144 MRPL37:313 RPS8:498 PABPC4:499 RPSA:502
One−Carbon Compound Transport (GO:001975	0.19809892	29	2.233e−04	5.013e−02	SLC4A1:127 SLC4A3:157 SLC4A7:357 SLC14A2:408 CA2:472 UPK3A:606
Cilium Movement (GO:0003341)	−0.15520117	47	2.343e−04	5.050e−02	CCDC40:36 DNAI2:49 CFAP251:178 TEKT1:449 CFAP54:564 RSPH6A:649
piRNA Processing (GO:0034587)	0.24576301	18	3.071e−04	6.364e−02	GPAT2:75 ANKRD34B:93 TDRKH:367 PNLDC1:535 FKBP6:697 DDX4:773
Aerobic Electron Transport Chain (GO:001	−0.14055474	54	3.568e−04	6.866e−02	NDUFB10:273 NDUFV3:343 NDUFA8:467 NDUFB8:482 NDUFS5:745 CYC1:896
Fatty Acid Catabolic Process (GO:0009062	0.13567636	58	3.556e−04	6.866e−02	ECI2:51 SLC27A4:52 LPIN2:202 DECR1:210 ABCD3:397 ACOT8:567
Cellular Response To Cytokine Stimulus (	−0.06778700	231	4.063e−04	7.549e−02	CSF2RB:18 BRCA1:46 CCL1:53 IL6ST:141 IL4R:204 CD40:207
DNA−templated DNA Replication Maintenanc	−0.16276361	38	5.199e−04	8.754e−02	ATRX:184 SLFN11:588 CENPS:1085 PRIMPOL:1164 MUS81:1369 TIMELESS:1484
Potassium Ion Transport (GO:0006813)	0.09269285	119	4.902e−04	8.754e−02	KCNK18:32 KCNJ16:53 KCNA4:111 KCNQ5:131 LRRC55:204 KCNS2:235
Regulation Of Type II Interferon Product	−0.11707396	74	5.046e−04	8.754e−02	PTPN22:225 IL18:356 RIPK3:466 CD14:491 TLR3:587 CD244:696
Lipid Transport (GO:0006869)	0.10340843	93	5.783e−04	9.164e−02	SLC27A4:52 ABCA5:64 ATP8B1:130 CD36:159 SLC10A2:245 MFSD2A:342
Regulation Of Cilium Beat Frequency (GO:	−0.44527553	5	5.642e−04	9.164e−02	CCDC40:36 CYB5D1:501 CFAP43:893 CFAP206:907 DNAH11:2291 NA
Gamma−Aminobutyric Acid Transport (GO:00	0.40167818	6	6.558e−04	9.920e−02	SLC6A6:123 SLC6A1:753 SLC6A12:947 SLC6A11:1901 SLC6A13:2738 SLC9A3R1:2875
Macromolecule Biosynthetic Process (GO:0	−0.07836913	159	6.704e−04	9.920e−02	MRPL24:144 MRPL37:313 RPS8:498 PABPC4:499 RPSA:502 HARS1:767
Nitric Oxide Mediated Signal Transductio	0.23801042	17	6.812e−04	9.920e−02	CD36:159 GUCY1A1:312 KCNC2:585 AGTR2:796 INS:942 NEUROD1:961
Mitochondrial RNA Processing (GO:0000963	−0.32617598	9	7.031e−04	9.969e−02	FASTKD5:469 TRMT10C:1049 TBRG4:1124 SUPV3L1:1239 FASTKD2:1896 PNPT1:1950
Replication Fork Processing (GO:0031297)	−0.15191560	41	7.681e−04	1.061e−01	BRCA1:46 BRCA2:140 ATRX:184 CENPS:1085 PRIMPOL:1164 MUS81:1369
B Cell Mediated Immunity (GO:0019724)	−0.25504853	14	9.537e−04	1.257e−01	CSF2RB:18 IL9R:567 CD70:1082 FCGR2B:1830 MSH2:2186 TLR8:2329

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.19766572	101	6.926e−12	4.494e−08	SLC1A1:47 SLC43A1:48 SLC6A6:123 SLC4A1:127 SLC4A3:157 SLC6A15:190
BENPORATH_ES_WITH_H3K27ME3	0.05183771	982	5.073e−08	1.097e−04	RGS9:26 DUSP8:43 LYSMD2:69 PADI2:97 KCNQ5:131 SNTB1:139
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.10335869	236	4.792e−08	1.097e−04	SLC1A1:47 SLC43A1:48 SLC27A4:52 SLC6A6:123 SLC4A1:127 SLC22A4:152
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	−0.20804483	47	8.087e−07	1.312e−03	NDUFB10:273 NDUFV3:343 TMEM126B:418 NDUFA8:467 NDUFB8:482 TMEM70:554
REACTOME_THE_CITRIC_ACID_TCA_CYCLE_AND_R	−0.11589499	149	1.076e−06	1.396e−03	ATP5PD:95 LRPPRC:153 NDUFA11:162 LDHC:168 ME2:254 NDUFB10:273
REACTOME_SLC_TRANSPORTER_DISORDERS	0.14663175	90	1.549e−06	1.674e−03	NUP160:12 SLC1A1:47 SLC27A4:52 SLC4A1:127 NUP210:185 SLC40A1:252
REACTOME_COMPLEX_I_BIOGENESIS	−0.19913763	48	1.825e−06	1.692e−03	NDUFA11:162 NDUFB10:273 NDUFV3:343 TMEM126B:418 NDUFA8:467 NDUFB8:482
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	0.06420633	420	6.897e−06	5.593e−03	FAR2:4 PODXL2:28 SKOR1:60 ANK1:141 MYO16:207 COL8A2:208
REACTOME_BETA_DEFENSINS	−0.32062540	16	8.983e−06	5.828e−03	TLR1:118 DEFB132:147 DEFB136:210 CCR2:940 DEFB129:953 CCR6:1205
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	−0.12846701	101	8.276e−06	5.828e−03	ATP5PD:95 LRPPRC:153 NDUFA11:162 NDUFB10:273 NDUFV3:343 TMEM126B:418
REACTOME_CHROMOSOME_MAINTENANCE	−0.12835939	99	1.033e−05	6.093e−03	CENPU:86 ATRX:184 POLR2H:285 MIS18BP1:514 CENPC:619 HJURP:636
REACTOME_DEFENSINS	−0.28949042	19	1.251e−05	6.766e−03	TLR1:118 DEFB132:147 DEFB136:210 CCR2:940 DEFB129:953 CCR6:1205
BENPORATH_PRC2_TARGETS	0.05266840	580	1.617e−05	6.994e−03	LYSMD2:69 FFAR4:156 PLXNA2:228 GHSR:231 RIMBP3B:248 TBX21:259
REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE	0.22483861	31	1.477e−05	6.994e−03	SLC43A1:48 SLC6A6:123 SLC6A15:190 SLC36A2:369 SLC6A19:389 SLC6A14:604
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	−0.13882683	81	1.582e−05	6.994e−03	LRPPRC:153 NDUFA11:162 NDUFB10:273 NDUFV3:343 TMEM126B:418 NDUFA8:467
FLORIO_HUMAN_NEOCORTEX	−0.34411677	12	3.664e−05	1.398e−02	FCRL4:104 FUT6:124 ACSM2B:459 FAM177B:519 PIWIL3:909 ERAP2:1160
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.06622272	331	3.637e−05	1.398e−02	ST6GAL2:80 ANKRD34B:93 FFAR4:156 FOXF2:175 GHSR:231 KCNS2:235
WP_OXIDATIVE_PHOSPHORYLATION	−0.17493354	46	4.063e−05	1.464e−02	ATP5PD:95 NDUFA11:162 NDUFB10:273 NDUFV3:343 ATP5MG:452 NDUFA8:467
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.05023488	555	5.659e−05	1.680e−02	SKOR1:60 CRACR2B:86 PADI2:97 RUNDC3A:132 FFAR4:156 SPRN:189
REACTOME_SPERM_MOTILITY_AND_TAXES	−0.41133721	8	5.599e−05	1.680e−02	CATSPERB:26 CATSPER3:241 CATSPER4:525 CATSPER2:935 CATSPERG:976 CATSPERD:1058
REACTOME_REPRODUCTION	−0.11834391	97	5.697e−05	1.680e−02	CATSPERB:26 BRCA1:46 BRCA2:140 CATSPER3:241 MLH3:414 ADAM21:481
REACTOME_DEPOSITION_OF_NEW_CENPA_CONTAIN	−0.18452025	40	5.405e−05	1.680e−02	CENPU:86 MIS18BP1:514 CENPC:619 HJURP:636 H2AC14:985 CENPS:1085
REACTOME_NA_CL_DEPENDENT_NEUROTRANSMITTE	0.26166365	19	7.863e−05	2.218e−02	SLC6A6:123 SLC6A15:190 SLC22A1:300 SLC6A19:389 SLC6A14:604 SLC6A1:753
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	−0.04646528	620	8.563e−05	2.315e−02	BIRC3:3 CSF2RB:18 TNFRSF6B:21 GH2:37 P4HB:48 LCK:51
SMID_BREAST_CANCER_BASAL_DN	0.04609402	627	8.940e−05	2.320e−02	LDLRAD4:3 ALB:20 SLC1A1:47 TSPAN1:78 MZF1:96 TIMP4:158
WEBER_METHYLATED_HCP_IN_SPERM_DN	0.22102907	26	9.587e−05	2.392e−02	GTF2A1L:496 SOX30:578 TULP2:591 STK31:671 DMRT1:724 RBMXL2:734
REACTOME_POSITIVE_EPIGENETIC_REGULATION_	−0.14364199	61	1.050e−04	2.434e−02	CHD3:122 GATAD2B:233 POLR2H:285 TBP:353 POLR1E:681 MYO1C:690
REACTOME_METABOLISM_OF_LIPIDS	0.04390027	680	1.049e−04	2.434e−02	FAR2:4 ALB:20 MORC2:42 ECI2:51 GPAT2:75 MOGAT2:81
REACTOME_ANTIMICROBIAL_PEPTIDES	−0.17335554	41	1.230e−04	2.660e−02	TLR1:118 DEFB132:147 DEFB136:210 RNASE8:583 ELANE:883 CCR2:940
REACTOME_INHIBITION_OF_DNA_RECOMBINATION	−0.19338070	33	1.210e−04	2.660e−02	ATRX:184 POLR2H:285 TERF1:798 POLR2G:901 TINF2:983 H2AC14:985
BENPORATH_EED_TARGETS	0.03727587	927	1.368e−04	2.863e−02	PARP12:49 LYSMD2:69 KCNQ5:131 FFAR4:156 FOXF2:175 PLXNA2:228
NIKOLSKY_BREAST_CANCER_5P15_AMPLICON	0.23082838	22	1.784e−04	3.528e−02	SLC6A19:389 LPCAT1:401 CEP72:763 AHRR:1137 SDHA:1193 EXOC3:1339
REACTOME_FERTILIZATION	−0.23605572	21	1.806e−04	3.528e−02	CATSPERB:26 CATSPER3:241 ADAM21:481 CATSPER4:525 CATSPER2:935 CATSPERG:976
VANTVEER_BREAST_CANCER_ESR1_UP	0.09261758	137	1.849e−04	3.528e−02	NPDC1:181 IGFBP4:217 ESR1:233 LONRF2:246 FARP2:278 ERBB4:359
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.04261867	667	1.906e−04	3.534e−02	ATP6V0A2:14 ATP10A:18 ALB:20 ATP2A1:29 SLC1A1:47 SLC43A1:48
REACTOME_RNA_POLYMERASE_I_PROMOTER_ESCAP	−0.15837441	46	2.029e−04	3.656e−02	POLR2H:285 TBP:353 POLR1E:681 H2AC14:985 POLR2F:1303 RRN3:1597
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	0.17370800	38	2.116e−04	3.710e−02	BRAT1:90 SUN1:386 TMEM184A:474 FBXL18:693 ADAP1:1207 UNCX:1385
REACTOME_RECOGNITION_AND_ASSOCIATION_OF_	−0.21156542	25	2.508e−04	4.240e−02	TERF1:798 TINF2:983 H2AC14:985 ACD:1051 NEIL3:1052 H2BC11:1915
REACTOME_AMINE_LIGAND_BINDING_RECEPTORS	0.16510588	41	2.548e−04	4.240e−02	HRH1:398 HTR6:418 ADRB1:490 GPR143:681 TAAR1:824 HTR1D:1144
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	0.04077753	698	2.641e−04	4.283e−02	HIRA:10 CTNND1:11 NT5C3B:30 SLC1A1:47 ABI2:62 ST6GAL2:80

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Mitochondrial Diseases	−0.07015658	347	7.986e−06	7.835e−02	ATAD3B:136 LRPPRC:153 NDUFA11:162 C3:248 ASPM:252 NDUFB10:273
Hypercalciuria	0.17963188	44	3.782e−05	1.579e−01	CLDN16:65 KCNJ1:191 TRPV6:196 PTH:234 PRPF4B:433 CLCN5:440
Nephrocalcinosis	0.15731859	54	6.439e−05	1.579e−01	CLDN16:65 SLC4A1:127 SDS:148 KCNJ1:191 PTH:234 HOGA1:318
Schizophrenia	0.03074998	1620	6.111e−05	1.579e−01	CNTF:6 TTC9C:9 RGS9:26 SLC1A1:47 TMX2:82 PADI2:97
Sterility, Postpartum	0.21942834	24	1.991e−04	1.775e−01	ESR1:233 LEP:413 ESR2:1121 LEPR:1157 NR1I2:1175 ADIPOR2:1352
Subfertility, Female	0.21942834	24	1.991e−04	1.775e−01	ESR1:233 LEP:413 ESR2:1121 LEPR:1157 NR1I2:1175 ADIPOR2:1352
Female infertility	0.20728545	28	1.475e−04	1.775e−01	STAG3:85 ESR1:233 LEP:413 ESR2:1121 LEPR:1157 NR1I2:1175
Female sterility	0.21942834	24	1.991e−04	1.775e−01	ESR1:233 LEP:413 ESR2:1121 LEPR:1157 NR1I2:1175 ADIPOR2:1352
Increased CSF lactate	−0.14847529	53	1.866e−04	1.775e−01	LRPPRC:153 NDUFA11:162 TMEM126B:418 NDUFA12:670 TRMT10C:1049 NDUFS7:1569
Paranasal Sinus Diseases	−0.17353929	39	1.780e−04	1.775e−01	CCDC40:36 PTPN22:225 DRC1:477 FMR1:699 ATM:701 SPAG1:795
Tetany	0.24035840	21	1.377e−04	1.775e−01	HIRA:10 CLDN16:65 KCNJ1:191 PTH:234 ARVCF:326 SLC12A3:571
Metabolic Diseases	0.05651149	361	2.456e−04	2.008e−01	CNTF:6 MOGAT2:81 ENPP2:84 GCDH:149 CD36:159 MAT1A:160
Bardet−Biedl Syndrome	0.14011358	55	3.285e−04	2.171e−01	VEGFA:375 LEP:413 LZTFL1:574 IQCB1:622 AZIN1:852 TMEM67:1364
Hyperactive renin−angiotensin system	0.36465537	8	3.548e−04	2.171e−01	KCNJ1:191 SCNN1G:260 SCNN1B:709 SLC12A1:793 CLCNKB:2448 NR3C2:3042
Increased plasma renin activity	0.36465537	8	3.548e−04	2.171e−01	KCNJ1:191 SCNN1G:260 SCNN1B:709 SLC12A1:793 CLCNKB:2448 NR3C2:3042
Pediatric Obesity	0.12940092	65	3.125e−04	2.171e−01	AMY1C:366 VEGFA:375 LEP:413 ACAD8:429 GHRL:673 MCHR2:754
Sinusitis	−0.14130860	53	3.763e−04	2.171e−01	PIP:7 CCDC40:36 PTPN22:225 DRC1:477 FMR1:699 ATM:701
Retinal Diseases	0.05829676	314	4.086e−04	2.227e−01	CNTF:6 ALB:20 RGS9:26 SLC6A6:123 CASP3:171 RD3:183
Lactic acidemia	−0.09270763	121	4.383e−04	2.263e−01	LRPPRC:153 NDUFA11:162 LYRM7:352 PPM1B:387 TMEM126B:418 SFXN4:503
22q11 Deletion Syndrome	0.23151878	19	4.774e−04	2.342e−01	HIRA:10 ARVCF:326 DGCR8:504 SNAI1:1102 COMT:1806 SEC24C:1918
Dyslipidemias	0.07299944	188	5.780e−04	2.700e−01	ALB:20 ENPP2:84 CD36:159 ESR1:233 PTGS2:399 LEP:413
Anemia, hereditary spherocytic hemolytic	0.43561218	5	7.423e−04	2.776e−01	SPTA1:33 SLC4A1:127 ANK1:141 SPTB:220 EPB42:4658 NA
DEAFNESS, AUTOSOMAL RECESSIVE (disorder)	0.16789874	34	7.072e−04	2.776e−01	WHRN:794 GJB6:914 OTOA:1530 CDH23:1628 GJA1:1769 PDZD7:1889
22q11 partial monosomy syndrome	0.29621412	11	6.698e−04	2.776e−01	HIRA:10 ARVCF:326 PI4KA:931 COMT:1806 SEC24C:1918 RREB1:2475
Glaucoma	0.05418270	330	7.640e−04	2.776e−01	CNTF:6 PADI2:97 CASP3:171 PROM1:187 NCK2:316 BAK1:335
Kidney Calculi	0.13380940	53	7.586e−04	2.776e−01	CLDN16:65 SLC4A1:127 TRPM7:128 PTH:234 LRPAP1:354 CLCN5:440
Spherocytosis	0.43561218	5	7.423e−04	2.776e−01	SPTA1:33 SLC4A1:127 ANK1:141 SPTB:220 EPB42:4658 NA
Bipolar Disorder	0.03698978	722	8.229e−04	2.805e−01	SLC1A1:47 PER2:106 CREM:142 CD36:159 PROM1:187 TRPV6:196
Idiopathic hypercalciuria	0.36484626	7	8.293e−04	2.805e−01	CLCN5:440 SLC12A3:571 SLC12A1:793 CASR:856 VDR:2332 TRPV5:3839
Metabolic Bone Disorder	0.17554033	30	8.790e−04	2.874e−01	ESR1:233 PTH:234 VEGFA:375 LEP:413 CA2:472 PHEX:491
Olivopontocerebellar Atrophies	0.21921229	19	9.415e−04	2.979e−01	SLC1A1:47 CASP3:171 GRIA2:639 ERCC6:698 GRIA3:703 GRM3:751
Acidosis, Lactic	−0.07760095	151	1.024e−03	3.053e−01	LRPPRC:153 NDUFA11:162 LYRM7:352 PPM1B:387 TMEM126B:418 SFXN4:503
Hypertensive disease	0.02980695	1106	1.027e−03	3.053e−01	ALB:20 SARDH:27 CLDN16:65 S100A6:95 SLC4A1:127 TRPM7:128
Hematologic Neoplasms	−0.04463708	465	1.062e−03	3.065e−01	BRCA1:46 ARR3:102 MTX1:123 BRCA2:140 ATRX:184 CD40:207
Hypokalemia	0.16313831	33	1.186e−03	3.325e−01	SLC4A1:127 KCNJ1:191 SCNN1G:260 SLC12A3:571 SCNN1B:709 SLC12A1:793
Asymmetric crying face association	0.25888885	13	1.230e−03	3.353e−01	HIRA:10 CASP3:171 ARVCF:326 COMT:1806 SEC24C:1918 RREB1:2475
Atherogenesis	0.14044668	44	1.275e−03	3.380e−01	ESR1:233 NPC1:310 ALOX5:340 VEGFA:375 HRH1:398 PTGS2:399
Leigh Disease	−0.10381095	80	1.344e−03	3.471e−01	LRPPRC:153 GHR:510 NDUFA12:670 DNAH8:726 LIAS:1100 EMC3:1525
Nicotinamide adenine dinucleotide coenzy	−0.14766858	39	1.425e−03	3.584e−01	NDUFA11:162 HSPB3:473 NDUFA12:670 PDSS1:1165 NDUFS7:1569 NDUFA10:1903
Fibrous bands	0.40994406	5	1.500e−03	3.674e−01	SARDH:27 SDS:148 BAX:361 BCL2:1609 SDHB:5077 NA

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0000716 abnormal immune system	−0.04962780	973	1.229e−06	5.850e−04	BIRC3:3 CSF2RB:18 CD207:25 SH2D2A:30 MOG:39 BRCA1:46
MP0002429 abnormal blood cell	−0.04341563	1191	4.719e−06	1.123e−03	BIRC3:3 CSF2RB:18 CD207:25 SH2D2A:30 MOG:39 BRCA1:46
MP0001765 abnormal ion homeostasis	0.12481436	103	1.391e−05	2.207e−03	CLDN16:65 SLC4A1:127 CYP11A1:180 KCNJ1:191 TRPV6:196 SCNN1G:260
MP0001800 abnormal humoral immune	−0.06672100	327	5.130e−05	5.342e−03	CFD:11 CSF2RB:18 LCK:51 NEDD4:54 C5AR1:96 GON4L:111
MP0000689 abnormal spleen morphology	−0.05299844	535	5.612e−05	5.342e−03	C5AR1:96 GON4L:111 TCIRG1:113 BRCA2:140 IL6ST:141 SGSH:170
MP0002452 abnormal antigen presenting	−0.05060420	566	8.023e−05	5.874e−03	BIRC3:3 CSF2RB:18 LCK:51 NEDD4:54 GON4L:111 TCIRG1:113
MP0002723 abnormal immune serum	−0.04830441	623	8.639e−05	5.874e−03	BIRC3:3 CSF2RB:18 SH2D2A:30 MOG:39 LCK:51 NEDD4:54
MP0005636 abnormal mineral homeostasis	0.09706706	134	1.215e−04	7.230e−03	ALB:20 CLDN16:65 SLC4A1:127 ANK1:141 CYP11A1:180 KCNJ1:191
MP0004043 abnormal pH regulation	0.29037151	14	1.710e−04	8.543e−03	KCNJ1:191 SLC4A7:357 WNK4:447 SCNN1B:709 SLC12A1:793 SOD2:1098
MP0001819 abnormal immune cell	−0.04001281	859	1.974e−04	8.543e−03	BIRC3:3 CSF2RB:18 SH2D2A:30 MOG:39 BRCA1:46 LCK:51
MP0005025 abnormal response to	−0.05995639	347	1.831e−04	8.543e−03	CFD:11 CSF2RB:18 CD207:25 C5AR1:96 BRCA2:140 IL6ST:141
MP0002420 abnormal adaptive immunity	−0.03706096	856	5.757e−04	2.284e−02	BIRC3:3 CSF2RB:18 SH2D2A:30 MOG:39 BRCA1:46 LCK:51
MP0000230 abnormal systemic arterial	0.08240848	143	7.583e−04	2.777e−02	GYS2:15 PTGS2:399 WNK4:447 CACNB3:477 PTGIS:481 NOX1:512
MP0003448 altered tumor morphology	−0.10570187	84	8.709e−04	2.961e−02	BRCA2:140 CXCR3:280 IQGAP1:289 TGM2:710 CXCR2:770 CHEK1:823
MP0009642 abnormal blood homeostasis	0.03086600	1250	9.334e−04	2.962e−02	GYS2:15 ALB:20 CLPS:24 ATP2A1:29 SLC27A4:52 FFAR1:61
MP0003948 abnormal gas homeostasis	0.05782280	284	1.021e−03	3.038e−02	ATP2A1:29 SLC27A4:52 MOGAT2:81 GRID2:198 GHSR:231 VEGFA:375
MP0009643 abnormal urine homeostasis	0.06306118	227	1.280e−03	3.583e−02	ALB:20 SLC1A1:47 CLDN16:65 SLC4A1:127 GCDH:149 KCNJ1:191
MP0002066 abnormal motor capabilities/c	0.03104402	1079	1.596e−03	3.844e−02	CNTF:6 SLC1A1:47 SLC27A4:52 ABCA5:64 PER2:106 ANK1:141
MP0002398 abnormal bone marrow	−0.03468718	794	1.780e−03	3.844e−02	SH2D2A:30 MOG:39 BRCA1:46 LCK:51 NEDD4:54 GON4L:111
MP0005387 immune system phenotype	−0.09208897	97	1.852e−03	3.844e−02	TCIRG1:113 C3:248 NFATC1:571 JAK3:693 AGER:841 TICAM2:862
MP0000685 abnormal immune system	−0.14568146	38	1.938e−03	3.844e−02	C3:248 LSP1:1262 SMAD3:1527 CMKLR1:1620 FAS:1757 ITGB7:1845
MP0001756 abnormal urination	0.10099458	81	1.781e−03	3.844e−02	ALB:20 SLC4A1:127 KCNJ1:191 TRPV6:196 SCNN1G:260 SLC14A2:408
MP0004019 abnormal vitamin homeostasis	0.14067502	41	1.883e−03	3.844e−02	CLDN16:65 TRPV6:196 PHEX:491 CYP7A1:812 SLC23A1:927 RBP2:996
MP0001790 abnormal immune system	−0.09208897	97	1.852e−03	3.844e−02	TCIRG1:113 C3:248 NFATC1:571 JAK3:693 AGER:841 TICAM2:862
MP0003656 abnormal erythrocyte physiolo	0.18947911	22	2.126e−03	4.048e−02	SLC4A1:127 ANK1:141 MAN2A1:330 KCNN4:652 STEAP3:1054 NFE2L2:1469
MP0002722 abnormal immune system	−0.03739609	605	2.692e−03	4.929e−02	CSF2RB:18 BRCA1:46 LCK:51 GON4L:111 TCIRG1:113 BRCA2:140
MP0003953 abnormal hormone level	0.03793996	559	3.276e−03	5.776e−02	GYS2:15 FFAR1:61 CLDN16:65 MOGAT2:81 CYP11A1:180 TRPV6:196
MP0002152 abnormal brain morphology	0.02897597	959	4.858e−03	8.258e−02	HIRA:10 CTNND1:11 ABI2:62 ENPP2:84 ANK1:141 LATS2:143
MP0004085 abnormal heartbeat	0.06332947	158	6.601e−03	9.241e−02	HIRA:10 ENPP2:84 SLC4A1:127 ERBB4:359 VEGFA:375 PTGS2:399
MP0002090 abnormal vision	0.12158664	42	6.555e−03	9.241e−02	SLC6A6:123 SLC4A3:157 SLC4A7:357 ELOVL4:460 MERTK:682 CYP7A1:812
MP0001486 abnormal startle reflex	0.08263978	95	5.690e−03	9.241e−02	LATS2:143 CASP3:171 MYO6:315 PCDH15:471 GRIA3:703 SLC6A1:753
MP0001666 abnormal nutrient absorption	0.12300393	42	5.953e−03	9.241e−02	ALB:20 CLPS:24 MOGAT2:81 TRPV6:196 PNLIPRP2:222 SLC10A2:245
MP0010329 abnormal lipoprotein level	0.08775658	82	6.306e−03	9.241e−02	CLPS:24 MOGAT2:81 CD36:159 ESR1:233 ALOX5:340 LEP:413
MP0002882 abnormal neuron morphology	0.02906480	880	6.328e−03	9.241e−02	CTNND1:11 SLC1A1:47 ABI2:62 ENPP2:84 SLC6A6:123 ATP8B1:130
MP0005365 abnormal bile salt	0.15140490	26	7.640e−03	9.828e−02	ALB:20 ATP8B1:130 SLC10A2:245 PNLIP:253 CYP7A1:812 HSD17B4:998
MP0001919 abnormal reproductive system	0.12233738	40	7.585e−03	9.828e−02	ESR1:233 BAX:361 PTGS2:399 WT1:427 CXCL12:550 LGR4:762
MP0006276 abnormal autonomic nervous	0.13926124	31	7.408e−03	9.828e−02	LEP:413 PRLH:674 AGTR2:796 LEPR:1157 GPSM1:1356 POU4F3:1586
MP0005360 urolithiasis	0.26876986	8	8.509e−03	1.066e−01	SLC26A1:536 SLC2A9:708 SLC3A1:2265 AGXT:2560 SLC7A9:2833 SLC26A6:3819
MP0000465 gastrointestinal hemorrhage	−0.15442440	24	8.935e−03	1.090e−01	IL6ST:141 IL18:356 FES:556 NFATC1:571 FOXP3:949 ACVRL1:1381
MP0002210 abnormal sex determination	0.04207766	331	1.022e−02	1.169e−01	PIWIL2:35 TAF7L:56 CREM:142 CYP11A1:180 RAD23B:200 ESR1:233

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
testis	−0.04639974	1382	8.069e−08	4.357e−06	DNAH14:2 SCNN1D:10 ZSCAN5B:19 C1orf100:23 SHBG:24 HEATR4:27
EBV.lymphocyte	−0.05210427	672	8.467e−06	2.286e−04	BIRC3:3 FCRL5:16 ICOSLG:32 BRCA1:46 LCK:51 POLQ:126
spleen	−0.06580039	299	1.144e−04	2.060e−03	FCRL3:12 FCRL5:16 FCRL6:195 FOLR3:198 CXCL6:242 TTC24:257
nucleus.accumbens	0.09799960	99	7.964e−04	1.075e−02	ANKRD34B:93 KCNA4:111 HTR6:418 FAT3:551 PENK:720 NWD1:827
cerebellum	0.04649085	376	2.371e−03	2.561e−02	SKOR1:60 MTBP:71 ESPNL:73 PPFIA4:103 RUNDC3A:132 MTCL1:167
bladder	0.16926059	24	4.138e−03	3.725e−02	PADI3:58 RD3:183 PSCA:500 UPK3A:606 PLA2G4A:782 CYP4B1:1040
Brodmann.area.9	0.05988798	182	5.733e−03	4.423e−02	KCNQ5:131 C3orf80:232 KCNS2:235 NCAM2:362 RXFP1:508 KCNC2:585
putamen	0.19814269	15	7.922e−03	5.347e−02	RGS9:26 KCNH4:242 SLC18A3:557 GRM3:751 ANO3:2482 OPALIN:2550
substantia.nigra	0.21622475	12	9.533e−03	5.720e−02	KCNE5:1147 FOXB1:1270 SLC6A3:1770 SLC6A11:1901 CHRM5:2077 DBX2:4121
thyroid	0.05685377	151	1.666e−02	8.994e−02	HIRA:10 KCNJ16:53 TMC4:133 FHDC1:239 PGF:291 SLC5A8:368
breast	0.17235088	15	2.090e−02	1.026e−01	LEP:413 ALX4:1079 CIDEC:1272 HCAR1:2192 LMX1B:2345 NPY2R:2548
cerebellar.hemisphere	0.03008270	509	2.348e−02	1.057e−01	PODXL2:28 MTBP:71 ESPNL:73 PPFIA4:103 RUNDC3A:132 ANK1:141
blood	−0.03740319	276	3.482e−02	1.447e−01	CSF2RB:18 SH2D2A:30 NFAM1:76 C5AR1:96 TLR1:118 DPEP2:145
esophagus.muscularis.mucosa	−0.10719924	26	5.877e−02	2.267e−01	PRUNE2:369 KCNMB1:799 COL4A6:1142 CCDC69:1358 CHRM3:1421 JPH2:1967
cortex.kidney	0.04742410	114	8.188e−02	2.764e−01	CLDN16:65 KCNJ1:191 SLCO4C1:307 OGDHL:328 SLC36A2:369 FABP1:438
pancreas	0.05243280	94	8.025e−02	2.764e−01	CLPS:24 PNLIPRP1:37 SLC43A1:48 TRPV6:196 PNLIPRP2:222 PNLIP:253
endocervix	0.05800105	72	8.986e−02	2.827e−01	ESR1:233 PRSS12:611 SP5:767 SNED1:837 IGFBP5:899 ADRA2C:1321
left.ventricle	−0.05886616	68	9.425e−02	2.827e−01	FSD2:307 ASB10:351 TNNI3K:460 HSPB3:473 S100A1:509 ASB18:697
ectocervix	0.09341954	25	1.063e−01	3.021e−01	ESR1:233 ADRA2A:1512 KRT34:1561 LRFN5:1624 RIPOR3:1857 FGF7:1944
atrium.auricle	−0.04632861	90	1.303e−01	3.249e−01	FSD2:307 ASB10:351 ADAMTS7:363 TNNI3K:460 HSPB3:473 S100A1:509
esophagogastric.junction	−0.15268333	8	1.349e−01	3.249e−01	CHRM3:1421 COL4A5:1514 ADCY5:6063 BARX1:6063 GADL1:6063 HOXA4:6063
ovary	0.04466649	102	1.208e−01	3.249e−01	FFAR1:61 SLC4A3:157 TMEM221:555 DLK1:636 BTNL2:871 STAR:941
prostate	−0.06958196	38	1.384e−01	3.249e−01	CPAMD8:17 CREB3L4:127 TTC6:196 SHROOM1:777 ANO7:797 SCUBE2:1036
subcutaneous.adipose	0.04616562	72	1.770e−01	3.983e−01	CD36:159 FABP9:334 TNMD:406 LEP:413 APOLD1:649 PPARG:902
C1.spinal.cord	0.02823772	182	1.927e−01	3.986e−01	ENPP2:84 PADI2:97 ELOVL7:146 PLCL1:302 SEC14L5:341 TNFRSF21:372
hypothalamus	0.04698062	66	1.882e−01	3.986e−01	LRRC55:204 PCDH15:471 FOXB1:1270 GRIN3A:1527 GLRA3:1570 B3GAT2:1716
liver	0.02094739	328	1.993e−01	3.986e−01	GYS2:15 ALB:20 SARDH:27 EXOC3L4:38 MOGAT2:81 SDS:148
cerebral.cortex	0.03371954	107	2.305e−01	4.445e−01	CAMK1G:584 DMTN:717 KCNF1:848 RNF208:901 LRTM2:1136 NPAS1:1337
transformed.skin.fibroblast	−0.01946437	259	2.868e−01	5.340e−01	NEDD4:54 P4HA3:78 DNAJB4:110 CXCL6:242 CNN2:245 RRAS2:250
greater.omentum	0.04983989	33	3.224e−01	5.804e−01	TIMP4:158 LEP:413 LPL:861 CIDEC:1272 RBP7:2212 IL6:2393
stomach	0.02925192	86	3.503e−01	6.102e−01	EPN3:216 PSCA:500 GPRC5C:561 GHRL:673 CLDN18:723 SLC9A4:772
adrenal.gland	0.02376635	108	3.959e−01	6.271e−01	SNTB1:139 PLEK2:177 CYP11A1:180 SLC40A1:252 MAP3K15:256 SLC14A2:408
esophagus.mucosa	0.01600893	220	4.181e−01	6.271e−01	PADI3:58 KRT32:112 EPS8L1:213 ADGRF1:333 MAL2:461 CALML3:473
fallopian.tube	−0.03354261	52	4.038e−01	6.271e−01	TCF23:419 GATA5:520 ANKRD66:679 BCHE:1022 TUBA4B:1338 MSLNL:1549
peyers.patch	0.01863733	167	4.097e−01	6.271e−01	MOGAT2:81 CCR9:240 SLC10A2:245 SLC6A19:389 KCNJ13:395 FABP1:438
vagina	0.02583641	91	3.963e−01	6.271e−01	ESR1:233 SCNN1G:260 CALML3:473 PSCA:500 TMEM40:722 CERS3:805
skeletal.muscle	−0.01314102	239	4.890e−01	7.137e−01	ALPK3:34 MSS51:270 FSD2:307 ART3:310 ASB10:351 TACC2:402
pituitary.gland	0.01250058	213	5.336e−01	7.583e−01	CFAP47:147 FFAR4:156 RD3:183 GHSR:231 RADX:449 ELAPOR1:519
caudate.nucleus	0.02706473	26	6.333e−01	8.768e−01	GPR37L1:186 GRM3:751 MAPK4:1067 MLC1:1937 ANO3:2482 GPR88:4920
minor.salivary.gland	−0.01436158	69	6.809e−01	9.192e−01	PIP:7 CRISP3:83 FUT2:169 MMP10:635 MMP27:825 DNASE2B:1393